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TECHNICAL PRESENTATION

CSA1518

Identity and Access Management (IAM) in Cloud Environments

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INTRODUCTION

- Identity and Access Management (IAM) controls who can access cloud resources.
- It verifies user identity before granting access.
- IAM improves cloud security and protects sensitive data.
- It supports secure login, user management, and access control.
- It is widely used in cloud platforms such as AWS, Microsoft Azure, and Google Cloud.



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PROBLEM STATEMENT

- Unauthorized users may access cloud resources.
- Weak passwords increase security risks.
- Managing multiple user accounts is difficult.
- Data breaches can occur due to improper access control.
- Organizations need a secure and centralized access management system.



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OBJECTIVES

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RESULT

- Secure authentication and authorization implemented.
- Role-Based Access Control (RBAC) improves access management.
- Multi-Factor Authentication enhances security. Reduced unauthorized access attempts.
- Better monitoring of user activities.
- Improved protection of cloud data and services.



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FUTURE IMPLEMENTATION

- Integrate AI-based threat detection.
- Support passwordless authentication.
- Enhance biometric authentication methods.
- Improve Zero Trust security implementation.
- Enable cross-cloud identity management.
- Automate user access management using AI.



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CONCLUSION

- IAM is essential for securing cloud environments.
- It ensures only authorized users can access resources.
- Authentication and authorization improve data protection.
- Role-based access simplifies user management.
- IAM helps organizations maintain security, compliance, and efficient cloud operations

